

Teams Qualification

Technical Document Specification



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Background

South Africa heads to the polls for Local Government Elections on 4 November 2026. The Electoral Commission of South Africa (IEC) and related government portals publish extensive data on voter registration, election results, party registration, and ward representation, but this data is spread across multiple systems, in a mix of live dashboards, downloadable files, and geographic atlases, rarely combined into a single coherent picture.

This qualification challenges you to turn that raw electoral data into insight.

Essential Skills and Knowledge

The following skills and knowledge are required:

- Proficiency in pre-processing, involving the preparation and cleansing of data.
- Expertise in Statistical Analysis to extract meaningful insights from datasets.
- Adeptness in Data Visualization for effective communication of findings.
- Competence in Machine Learning to build a predictive model.
- Proficiency in developing data deployment tool for the model.

Participants have the flexibility to utilize any language, framework, development environment, library, or API in their solution.

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Teams Qualification Description

Using one or more of the datasets below, build a solution that helps voters, civil society, journalists, or policymakers better understand voter registration, electoral participation, or representation ahead of and around the 2026 Local Government Elections.

You have full freedom to define your own specific problem statement within this theme. We're not looking for one "correct" angle, we want to see how you frame a real question and answer it with data. Strong submissions typically:

- Identify a specific, well scoped question (not "analyze the election" but "which municipalities show the widest gap between registered youth and registered older voters, and has that gap widened since the last election?")
- Justify why the question matters and to whom (voters, IEC, parties, journalists, researchers)
- Combine or cross reference more than one dataset where possible
- Present a clear, honest account of data limitations

Dataset Sources

1. IEC Voter Registration Statistics page

<https://www.elections.org.za/pw/StatsData/Voter-Registration-Statistics>

Live, filterable dashboard of registered voters by province, municipality, age group, and gender.

- *Note: this is a dashboard, not a direct file download, you'll need to extract data per province/municipality selection (manually or via scraping) to build a structured dataset.*

2. IEC Results & Downloadable Election Results

<https://results.elections.org.za>

- **Municipal local government elections dashboard + downloadable results:** <https://results.elections.org.za/home/downloads/me-results>
- **Municipal by-elections dashboard + downloads:** <https://results.elections.org.za/home/downloads/mbe-results>

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- **National/provincial downloadable results:**
<https://results.elections.org.za/home/downloads/npe-results>
- **Atlas of (municipal) results — geographic/visual:**
<https://atlas.elections.org.za/lgeatlas/>
- **Atlas of (national/provincial) results:**
<https://atlas.elections.org.za/npeatlas/>

3. Party registration statistics

<https://www.elections.org.za/pw/StatsData/Political-Parties-Statistics>

Numbers and details of political parties registered to contest elections.

4. Ward councilor statistics

<https://www.elections.org.za/pw/StatsData/List-Of-Current-Ward-Councillors>

Current ward councilors by ward: a baseline of representation before new results come in.

Note: *The datasets listed above are your recommended starting points, but you are welcome to explore and incorporate additional publicly available datasets to support your problem statement for example, census, demographic, or socioeconomic data that could add context to your analysis. Any supplementary data should be clearly cited and its source documented in your submission.*

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Deliverables

You are required to submit:

- A defined problem statement, the specific question you're addressing within the election/civic participation theme, and why it matters
- A working notebook/codebase covering your full pipeline (data collection through model output)
- A trained model (predictive or otherwise) that directly addresses your problem statement
- A short video submission demonstrating your solution and key findings
- Presentation slides summarizing your approach, results, and recommendations

Teams Qualification Marking Rubric

Data understanding and exploration

- Data cleaning and processing (e.g. handling missing values, inconsistent formats, merging datasets from multiple sources)
- Exploratory data analysis and visualization
- Feature engineering (e.g., deriving turnout rates, registration gaps, fragmentation indices, or geographic features from the raw IEC data)

Model

- Model selection, training, testing and evaluation
- Model should directly address the problem statement

Note: should you choose a predictive model, be mindful that some of the provided sources are largely current snapshot data. A strong predictive model generally needs historical/time series data to learn from trends over time, so consider combining snapshot data with historical data where relevant.

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Analysis, results and interpretation

- Clear interpretation of model outputs in relation to the original problem statement
- Honest discussion of data limitations (e.g., dashboard-only data requiring extraction, incomplete historical coverage) and how these were addressed or mitigated

Deployment of model

- Model/s should be deployed in a usable form (e.g., an app, dashboard, API endpoint, or interactive notebook) rather than existing only as a static script

Code documentation

- Organization and quality of code documentation
- Notebooks should be clear, reproducible, and well-commented

Presentation

- Video submission
- Presentation slides

Bonus points

- Innovative techniques (e.g., creative feature engineering from geographic atlas data, novel cross-referencing of registration and results data, or unconventional but well-justified modeling approaches)

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Challenge 1 Submission

- You are required to submit your code by **8am on 28th September 2026**
- You are required to record a 15-minute presentation of the work you have done each member of the team needs to present.
- Your submission must be done on the NextCloud platform
<https://sdcdata.dirisa.ac.za/index.php/login>
- Only **one** team member needs to upload the relevant submission that comprises of:
 - Notebook with code, comments and references to packages used if needed.
 - Video recording:
 - Of the presentation of the solution (can be separated from the 15-minute demonstration)
 - Presentation slides with references.
- The team members need to create a shareable link on NextCloud. This link must be shared with Ms. Boitshepo Sebushe via email (bsebushe@csir.co.za).
- The challenge is only open to team members who received the email that they qualified.